

FreeIPA: The One-Stop Security Solution for SysAdmins



Systems administrators can now manage security in one centrally located place with FreeIPA, which handles authentication, authorisation and account information. They have the benefit of a CLI or a Web UI for management and administration of systems.

FreeIPA is an integrated security information management solution built using various open source technologies like 389 Directory Server, MIT Kerberos, NTP, DNS and the Dogtag Certificate System.

FreeIPA provides users and administrators a one-stop solution for integrated identity and authentication solutions in the Linux environment. Using FreeIPA server, systems administrators can manage authentication, authorisation and account information at a centralised place.

It provides both CLI and Web UI for management and administration.

The components of FreeIPA are:

1. 389 Directory Services: LDAP directory infrastructure
2. MIT Kerberos: For single-sign-on authentication
3. Dogtag: A certificate authority
4. Bind: A DNS service
5. SSSD: A client side component that integrates with FreeIPA for authentication

FreeIPA is a reliable solution as it is built on top of very well-known open source components and standard protocols. It is easy to install, manage and configure FreeIPA server and clients.

Getting started with FreeIPA

Installing the OS: Sysadmins can use Fedora or CentOS virtual machines for configuring and trying out FreeIPA in a

given environment. They can use public demo servers to get a quick idea about FreeIPA server, which can be found at <http://www.freeipa.org/page/Demo>

Setting up a static hostname: Kerberos authentication relies on a static hostname; so in order to satisfy this requirement, the host should either have a fully qualified domain name from the DNS server or a static hostname in the `/etc/hosts` file.

```
$ echo 10.10.10.1 ipaserver1.example.com ipaserver1 >> /etc/hosts
```

Here, we are using `ipaserver1.example.com` as the static hostname and assign `10.10.10.1` as the IP address to it in the `/etc/hosts` file, in order to get the static hostname resolved.

Installing FreeIPA server: We will use the standard FreeIPA server packages provided by the OS such as Fedora or CentOS. You can always choose to install from the upstream source hosted on <https://git.fedorahosted.org/cgi/freeipa.git>

Install the FreeIPA server by issuing the following command when using Fedora 21/CentOS 6:

```
# yum install freeipa-server
```

The installation log for `freeipa-server` is located at `/var/log/ipaserver-install.log`